

Abhay Deshpande (He/Him)

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Education

Princeton University, PhD in Electrical and Computer Engineering September 2026 – Present

- Working in the PRISM Lab with Prof. Dhruv Shah

University of Washington, BS in Computer Science and BA in Math June 2024

- GPA: 3.97/4.0
- Completed graduate-level coursework in: Machine Learning, Reinforcement Learning, and Robotics

Publications

- A. Deshpande*, M. Guru*, R. Hendrix*, S. Jauhri*, . . . , D. Shah[†], R. Krishna[†]. “MolmoBOT: Large-Scale Simulation Enables Zero-Shot Manipulation”. Preprint. (3x Workshop Best Paper @ ICRA2026)
- Y. Kim*, W. Pumacay*, O. Rayyan*, M. Argus*, . . . , A. Deshpande, . . . , R. Krishna. “MolmoSpaces: A Large-Scale Open Ecosystem for Robot Navigation and Manipulation”. RSS 2026.
- A. Deshpande, Y. Deng, A. Ray, J. Salvador, W. Han, J. Duan, K.H. Zeng, Y. Zhu, R. Krishna, R. Hendrix. “GraspMolmo: Generalizable Task-Oriented Grasping via Large-Scale Synthetic Data Generation”. CoRL 2025.
- A. Deshpande, L. Ke, Q. Pfiefer, A. Gupta, S. Srinivasa. “Data Efficient Behavior Cloning for Fine Manipulation via Continuity-based Corrective Labels”. IROS 2024 (Oral).
- L. Ke*, Y. Zhang*, A. Deshpande, S. Srinivasa, A. Gupta. “CCIL: Continuity-based Data Augmentation for Corrective Imitation Learning”. ICLR 2024.
- Y. Zhang*, L. Ke*, A. Deshpande, A. Gupta, S. Srinivasa. “Cherry Picking with Reinforcement Learning”. RSS 2023.

Experience

Predoctoral Researcher, PRIOR @ Allen Institute for AI September 2024 – June 2026

- Performed research in learning for robotic manipulation on a variety of real-world platforms
- Heavily utilized simulation for large-scale data collection and pretraining, including platforms like MuJoCo
- Published research papers on robotic manipulation in top robotics venues
- Leveraged large foundational VLMs and LLMs to apply modern AI to challenging robotic problems

Undergrad Researcher, Personal Robotics Lab + WEIRDLab @ UW June 2021 – June 2024

- Performed research in learning for robotic manipulation on real-world platforms, primarily focusing on a chopsticks robot for fine manipulation
- Leveraged data-driven control algorithms in Reinforcement Learning and Imitation Learning to create state-of-the-art robot policies that outperform humans and other baselines

Robotics Software Engineering Intern, NASA Jet Propulsion Laboratory June 2023 – September 2023

- Built and maintained tools that supported the planning process for the Curiosity Mars rover
- Automated target evaluations and safety checks for the rover’s arm, speeding up planning process by >10x

Activities

Titan Robotics Club, Robotics Mentor 2025

- Mentored for a high-school robotics club, teaching fundamentals of robotics and engineering
- Taught basics of classical robotic control and state estimation, including PID and basic kinematics

Husky Robotics Team @ UW, Software Lead 2020 – 2024

- Led the software subsystem, developing a full robot stack, from low-level control to high-level autonomy
- Championed inclusion and diversity, broadening outreach to new members and improving gender diversity